CSE221
Algorithm Analysis & Design

What will we learn in this course?

- Real life problems
- ***Naive solution*** vs ***efficient solution***

A very simple example of what is coming next



Bringing order to cards

- Suppose you want to keep the cards in either increasing or decreasing order.
- You are given 100 cards.
- Can you do it? How?

Sorting Algorithm

- You can solve most of the problems that you encounter in the course.
- Then why this course?
- Efficiency !!!!!
- Of course you could ‘sort’ the cards, but you need to do it in an efficient way.
- How to learn this efficient way?
- By learning algorithms and understanding how the algorithm works.

What will come in the exam question?

Problem:

A museum wants to **redesign the display of its ancient coins collection**. Each coin has a small tag with its **year of minting (which may include BCE and CE years) and a unique catalog ID.**

The curator wants the coins to be placed in a single glass case in such a way that a visitor walking from left to right can observe the smooth progression of time — **from the oldest coin to the newest.**

What will come in the exam question?

Problem:

A museum wants to **redesign the display of its ancient coins collection**. Each coin has a small tag with its **year of minting (which may include BCE and CE years) and a unique catalog ID.**

The curator wants the coins to be placed in a single glass case in such a way that a visitor walking from left to right can observe the smooth progression of time — from the oldest coin to the newest — **while ensuring that if two coins share the same year, they appear in the order of their catalog ID.**

How much do I need to write codes?

- Unlike DS course, you mostly have to write pseudo codes, steps, modification of some steps of the algorithm.
- BUT YOU NEED TO CODE IN LAB.

Quick Overview

- **Final - 30%**
- **Mid - 20%**
- **Assignment - 5% (Best 3 out of 4)**
- **Lab - 25%**
- **Quiz - 20% (Best 3 out of 4)**

Some Tips



Some Tips



Some tips

- Be confident.

Some tips

- Study regularly

Some tips

- Focus on the process, steps and intuition.

What will we do in class?

- Topic
- Slides
- Books
- Extra reading materials
- Type of qs

Any other question?